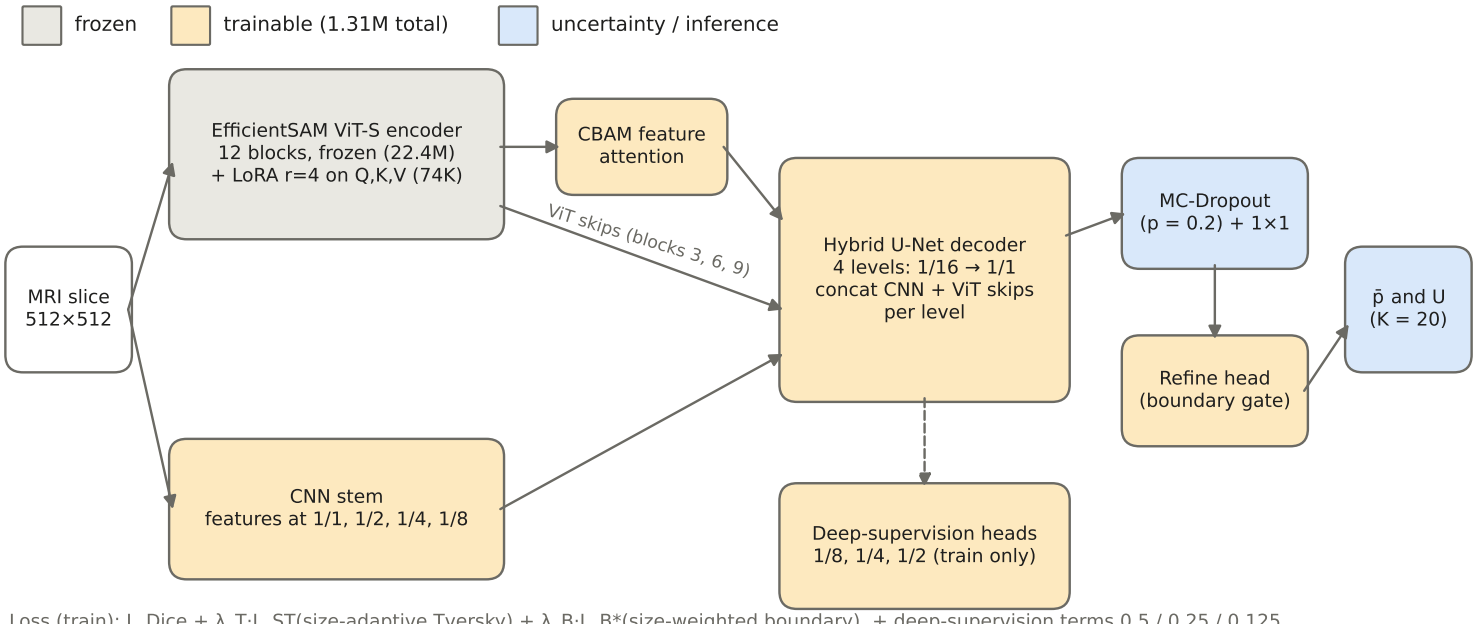
frozen
trainable (1.31M total)
uncertainty / inference
MRI slice
512×512
EfficientSAM ViT-S encoder
12 blocks, frozen (22.4M)
+ LoRA r=4 on Q,K,V (74K)
CBAM feature
attention
ViT skips (blocks 3, 6, 9)
CNN stem
features at 1/1, 1/2, 1/4, 1/8
Hybrid U-Net decoder
4 levels: 1/16 → 1/1
concat CNN + ViT skips
per level
Deep-supervision heads
1/8, 1/4, 1/2 (train only)
MC-Dropout
(p = 0.2) + 1×1
Refine head
(boundary gate)
p̄ and U
(K = 20)
Loss (train): L_Dice + λ_T·L_ST(size-adaptive Tversky) + λ_B·L_B*(size-weighted boundary), + deep-supervision terms 0.5 / 0.25 / 0.125